

Category: Productivity / Developer Tools**Target users:** Mobile AI API users.**Repository:** github.com/LiuZongrun7/Mobile_Group_20

1. APP NAME

TokenTrail

2. App Category

Productivity / Developer Tools.

3. About the App

TokenTrail is a mobile assistant for students and independent developers who pay for AI API calls across more than one provider. Their spending is often split between provider consoles, coding-agent logs and manually checked balance pages, so it is hard to answer simple questions such as: how many tokens did I use, why did my cost rise, how much balance may remain, and which price version was used for the estimate.

After a user registers or logs in, TokenTrail opens to a dashboard that summarises recent AI API usage, CNY estimated spending, monthly budget progress, data coverage and missing records. The app records or imports authorised pay-as-you-go usage, including input tokens, output tokens and provider-supported cache fields such as read cache and write cache. Each usage detail also carries lightweight data-source labels, so the user can see whether a value came from an API record, an imported log, a price table or manual input.

The main workflow is practical: view usage analytics, check cache and cost breakdowns, enter a manual AI wallet balance, forecast how many days the balance may last, and tap a small Home price icon when price or top-up information is needed. This Home price panel stores model price versions, official source links and official provider top-up links, while the Forum separates official updates from community tips. A home-screen AI assistant lets users type or dictate questions such as **"Why did this week cost more?"**; it answers only from TokenTrail summaries and keeps its own model cost in a separate ledger.

Scope is **pay-as-you-go API usage**. Token Plan, ChatGPT-style subscriptions and other quota plans are excluded because they cannot be treated as per-call API spending. The app estimates costs from saved usage and price records; it does not promise to recover every provider's full history, read hidden account balances or process payments. Tower-defence gameplay is an extension: verified usage can become game resources, but playing never changes real usage, budget or balance data.

4. Existing similar apps with source code and the app chosen to build upon

The review compares first-party billing views, open-source cost tools and agent plugins on data scope, source transparency and decision support.

Name	App Details	Source code link
OpenAI Platform usage	First-party usage and billing views for one OpenAI API account. TokenTrail adds a mobile cross-provider estimate view with coverage labels.	Docs.
Xiaomi MiMo API console	Provides model access and pay-as-you-go prices. TokenTrail keeps a manual/import fallback when detailed usage records are unavailable.	Docs.
DSH / dsh-context	DSH runs coding sessions; dsh-context visualises context, token and cache activity. TokenTrail imports usable records into Android without live agent control.	DSH; dsh-context.
LiteLLM	Open-source multi-provider gateway with spend tracking and budgets for proxied calls. TokenTrail can analyse imported or recorded usage without mandatory proxying.	GitHub.
AIUsage	Open-source local usage parser for AI tools; reads local logs and stores usage in a local database. TokenTrail adds an Android dashboard and gamified progress.	GitHub.

5. Main new added features and enhancements

Clear user flow. Login leads to a dashboard for usage, cost, budget and missing-data status. **Usage and cache analytics.** TokenTrail records input, output and available cache read/write tokens, then prices them with dated provider rates. **Manual AI wallet and runway.** Users enter provider balances manually; the app subtracts estimated spending and forecasts remaining days from recent usage. **Home price icon and top-up link.** A small Home icon opens model prices, effective dates, source links and official billing links without handling payments. **Evidence-based AI assistant.** Typed or confirmed speech questions are answered from read-only summaries, with the assistant's own API cost kept separate. **Forum and game extensions.** Official updates, user tips and tower-defence rewards support the analytics experience without changing financial records.

Scope rule: TokenTrail is first a mobile AI API usage and wallet-tracking app with a lightweight Home price panel. Data-source labels, forum posts and tower-defence gameplay support that core flow; model-quality recommendation, user rating, automatic hidden balance reading and payment processing are outside the first release.

6. Functional Requirements (App Features)

Ser	Existing Features	Proposed Improvements / New Features	Remarks
1	Account login	Register, log in/out, reset passwords and isolate usage, wallet, assistant, forum and game data by user.	Improved
2	Home dashboard	Show monthly tokens, estimated CNY cost, budget progress, data coverage and missing-record warnings after login.	Improved
3	Usage analytics	Track provider, model, date, calls, input tokens, output tokens, cache tokens and estimated cost by day/week/month.	Improved
4	Data-source labels	Small labels show whether usage, cost, cache and balance values are recorded, imported, estimated or manually entered.	New
5	Cache analytics	Display read cache, write cache and simple cache ratio inside the usage view when the provider exposes cache data.	Improved
6	Manual AI wallet	Let users enter and update provider balances manually; spending estimates reduce the displayed remaining balance.	New
7	Runway forecast	Estimate how many days a balance may last from recent average spending and warn when the runway is short.	New
8	Home price icon	A small Home icon opens model price versions, official source links and official billing/top-up links.	New
9	Forum / updates	Separate official provider updates from community tips, posts and replies; community tips are not pricing authority.	Improved
10	Home AI assistant	Type or dictate questions; assistant answers from read-only summaries and records its own API cost separately.	New
11	Tower defence	Convert verified usage into monthly game resources without changing real usage, balance or budget records.	New

7. Your Key contributions/ improvements

The main contribution is a complete mobile workflow from login to action: the user sees usage, understands the cost source, manages a manual balance, receives runway and budget warnings, checks price changes, asks the assistant for evidence, and can use verified usage in a game layer.

Account, dashboard and usage analytics.

Users register or log in with account isolation. The home dashboard shows recent tokens, estimated CNY spending, budget progress, coverage and missing records. Usage records include provider, model, timestamp, call count, input tokens, output tokens, cache read/write tokens when available, status and cost estimate. Daily, weekly and monthly summaries support trend views and provider/model comparisons.

Data-source labels and cache analytics.

TokenTrail does not hide uncertainty. Small labels mark values as recorded, imported, estimated, manually entered or unavailable. Input and output tokens are baseline fields; cache read/write fields are stored only when a provider or imported log exposes them. The cache view stays lightweight: it shows cache token totals and a simple cache ratio to help users see whether cached context may reduce repeated input cost.

Manual AI wallet and runway.

Because many API keys cannot read account balance, users enter provider balances manually and may update them at any time. TokenTrail subtracts estimated spending, shows remaining balance and forecasts runway from recent average daily cost. When balance or runway is low, the app shows an advisory warning and directs users to the Home price icon for official top-up links.

Home price icon and official updates.

The Home price icon opens a lightweight price panel instead of a separate page. It saves provider/model prices, token class, cache price, currency, effective date, source URL and official billing/top-up links. Historical estimates use the matching price version instead of applying today's price backward. Forum/Updates still separates official provider news from community posts.

Home AI assistant and separate cost tracking.

A floating "Ask AI / hold to talk" entry opens the home assistant. Users can type, or dictate and confirm the transcript before sending. The assistant reads only validated summaries such as usage, budgets, wallet runway, price versions, cache totals and forum highlights. Replies cite coverage and missing data, and the assistant's own model tokens/costs are stored in a separate ledger excluded from tracked provider totals and game rewards.

Tower-defence game.

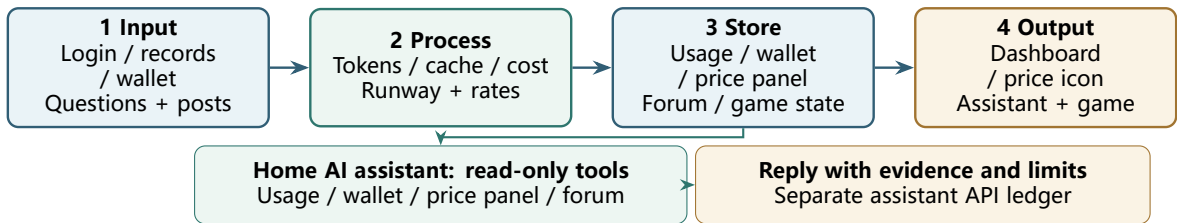
The game converts verified usage into monthly resources for towers, walls or upgrades. Re-importing the same

usage record grants no duplicate resources. Game actions never change real API usage, wallet balance, budget, price records or assistant cost; the purpose is motivation and retention, not financial calculation.

Data aspects.

Aspect	Planned implementation	User value / evidence
Data input	Account login, provider profiles, manual balances, budgets, app-recorded/imported usage, price records, text/confirmed speech, posts and game actions.	Covers the full app flow from login to wallet, assistant and game while keeping subscriptions excluded.
Data processing	De-duplicate records; calculate CNY costs, cache ratio, budget progress, wallet runway, price-panel warnings and read-only assistant summaries.	Turns token logs into actions: understand spending, decide whether to top up and ask evidence-based questions.
Data storage	Room stores local usage, prices and wallet state. Server store syncs accounts, budgets, forum, game state and separate assistant ledger.	User isolation is clear; financial records, assistant usage and game rewards do not overwrite each other.
Data output	Dashboard, trends, source labels, cache breakdowns, wallet/runway cards, Home price panel, top-up links, assistant answers, forum and game feedback.	Users see what happened, what is estimated, what is missing and what action to take next.

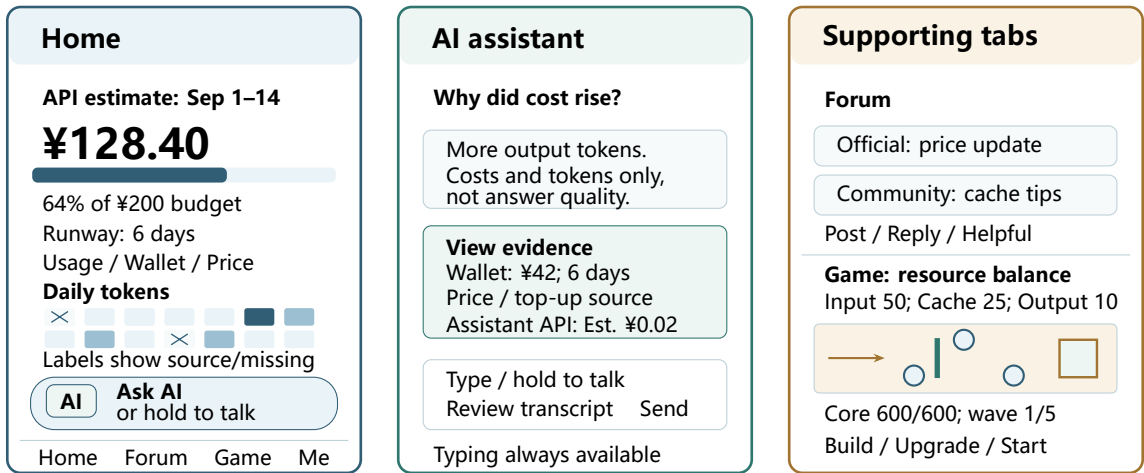
Data flow and limitations.



PAYG means pay-as-you-go. Verify source type, log version, billing mode and price version; some imported logs may require JSONL or Zstandard decoding. Read-only tools such as *getUsageSummary*, *getWalletRunway*, *getPriceVersions*, *getForumHighlights* and *getMyThreads* enforce the UID from the caller session and bounded queries. Models receive summaries, not secrets or raw prompts; missing records mean no personalised cost claim. No coding-agent control and no payment processing.

Proposed Android interface.

Home leads with analytics, wallet runway and the assistant entry. Tap to ask; hold to dictate, then confirm the transcript. Usage details expose source labels, token classes and cache values only where records support them. The Home price icon, Forum and Game are supporting entries. All sketches use illustrative September 2026 data; assistant API costs are excluded from the home total.



Interaction and accessibility.

Use labelled touch targets, text scaling and rotation support. Missing logs, rates and unsupported cache fields show *Unavailable*, not zero. Loading, error, low-balance, listening and permission-denied states are explicit; typing and analytics remain available if the model fails. The assistant never covers data or navigation.

8. Tool Stack

1. APIs that will be used.

OpenAI pricing and API documentation; Xiaomi MiMo pay-as-you-go pricing; DeepSeek pricing; Codex auth; ZCode configuration; DSH session logs; official billing/top-up URLs opened from the Home price panel through browser intent; OpenAI / Xiaomi MiMo / DeepSeek model APIs for the assistant service.

2. Tools that will be used.

Layer	Planned tools and rationale
Android interface	Java/XML, Material Components, Navigation and ViewModel/LiveData; RecyclerView lists, speech-to-text and typed fallback.
Network and work	Retrofit/OkHttp connects the Java advice service; bounded read-only tools and one model API. WorkManager checks prices, runway alerts and official update refreshes.
Persistence and security	Room stores usage, prices, manual wallet balances and history; the team server supports sync and public posts. The service authenticates every request and scopes every read to the account from the caller session; no client holds a model key.
Open-source reuse	Room (Apache-2.0): used for local records. MPAndroidChart (Apache-2.0): added dependency; charts planned. Room with a View (Apache-2.0): archived architecture reference, no copied code. dsh-context/LiteLLM: comparison only; dsh-pet: entry inspiration only. Game engine: original. Record reused files, versions, licences and art sources.

9. Development Timeline

Timeline for implementation.

Checkpoints are approximately Alpha (weeks 8–9), Beta (11–12) and Final (15–16); implementation and documentation are planned through week 15, with regular GitHub commits.

Course weeks	Work	Checkpoint / acceptance
1–3	Verify sample usage records, cache fields, provider prices, billing modes, Home price-panel top-up links, import formats and UI; define access rules.	Verified sources; isolated accounts.
4–7	Build import, storage, dated pricing, manual wallet, runway forecast and analytics; define read-only tool contracts.	Correct rates; idempotent import.
8–9: Alpha	Login → profile → import → cost → wallet → home.	Separate real/sample data; source, GitHub link and MP4.
10–12: Beta	Wang: assistant, Home price panel and forum using query interfaces from Zhang. Liu: settlement and real-time game.	Cited evidence; typing fallback; ranked posts; smooth waves.
13–15: Final	Validate wallet/runway warnings, price-panel top-up links, advice, cost separation, game balance, accessibility and isolation; complete documentation.	Demo, tests, report/video; licence records.

Team work and evidence of completion.

Zhang Li (24107757): authentication, usage record/import storage, dated pricing, budgets, dashboard and data-query interfaces.

Wang Tingdong (24107759): home-screen AI assistant, text/voice panel, Java advice service, five read-only tools and assistant API ledger; forum news, posts/replies and ranking.

Liu Zongrun (24107745): tower-defence engine/UI, resource settlement, monthly state, placement/upgrades and enemy waves.

All share UI/UX, accessibility and integration, and provide commits/module acceptance evidence.

Demo: Record or import usage → cost/session/budget views → ask “Why did cost rise?” by text/confirmed speech → evidence and separate assistant API cost. Then show forum posts and a player-started game wave.

Validation: Test plan exclusion, log formats, token classes, dated/time-based rates, missing data, de-duplication and account/tool isolation. Check replies against evidence, text fallback and ledger separation; game re-imports grant no duplicate resources. Sample data never enters real totals.